As we will see, superstring theory starts off by proposing a new answer to an old question: what are the smallest, indivisible constituents of matter? For many decades, the conventional answer has been that matter is composed of particles—electrons and quarks—that can be modeled as dots that are indivisible and that have no size and no internal structure. Conventional theory claims, and experiments confirm, that these particles combine in various ways to produce protons, neutrons, and the wide variety of atoms and molecules making up everything we've ever encountered. Superstring theory tells a different story. It does not deny the key role played by electrons, quarks, and the other particle species revealed by experiment, but it does claim that these particles are not dots. Instead, according to superstring theory, every particle is composed of a tiny filament of energy, some hundred billion billion times smaller than a single atomic nucleus (much smaller than we can currently probe), which is shaped like a little string. And just as a violin string can vibrate in different patterns, each of which produces a different musical tone, the filaments of superstring theory can also vibrate in different patterns. These vibrations, though, don't produce different musical notes; remarkably, the theory claims that they produce different particle properties. A tiny string vibrating in one pattern would have the mass and the electric charge of an electron; according to the theory, such a vibrating string would *be* what we have traditionally called an electron. A tiny string vibrating in a different pattern would have the requisite properties to identify it as a quark, a neutrino, or any other kind of particle. All species of particles are unified in superstring theory since each arises from a different vibrational pattern executed by the same underlying entity.

Going from dots to strings-so-small-they-look-like-dots might not seem like a terribly significant change in perspective. But it is. From such humble beginnings, superstring theory combines general relativity and quantum mechanics into a single, consistent theory, banishing the perniciously infinite probabilities afflicting previously attempted unions. And as if that weren't enough, superstring theory has revealed the breadth necessary to stitch all of nature's forces and all of matter into the same theoretical tapestry. In short, superstring theory is a prime candidate for Einstein's unified theory.